

CONSTRUCTION OS

PRD - IT Administration, Integrations & Service Management

Implementation-ready Product Requirements Document for Product, Enterprise Architecture, IT Operations, Backend, Frontend, Security, QA, DevOps and SRE Teams

Product purpose

Create the controlled company-level technical operating layer for identities, Microsoft 365, Outlook, Google Workspace, connected applications, devices, software, service support and security operations without expanding business-data access.

Document: PRD_IT_Administration_Integrations_Service_Management

Version: 1.0

Status: Implementation-ready functional and technical specification

Primary surfaces: Company IT workspace, service portal, mobile support and administration

Core architecture: Company IT domain + provider adapters + service management + controlled cross-module references

Access model: Deny by default; company scoped; field restricted; secrets isolated; fully audited

Date: 2026-08-06

ConstructionOS Confidential

Document Control

Field	Value
Document	PRD_IT_Administration_Integrations_Service_Management
Version	1.0
Status	Implementation-ready product, domain and technical specification
Primary owner	ConstructionOS Product and Platform Architecture
Business owners	Company Owner, IT Director and designated Security authority
Audience	Product, UX/UI, backend, frontend, integration, security, QA, DevOps, SRE and IT operations
Primary routes	/it, /it/integrations, /it/access, /it/devices, /it/service-desk
Source-of-truth	IT owns technical enablement, connector configuration, devices profiles, licences and service management
Integration posture	Provider-neutral adapters; Microsoft 365 and Google Workspace first; core remains self-contained
Date	2026-08-06

Permanent architecture rule

Company IT owns technical enablement, integrations, devices, licences and support. It does not automatically own or gain access to company business records.

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1. Executive Summary

The IT Administration, Integrations & Service Management Module is the controlled technical operating layer for each ConstructionOS company. It provides one governed workspace for identity-provider connections, Microsoft 365 and Outlook, Google Workspace, user provisioning, devices, licences, connected applications, service support, technical changes and security operations.

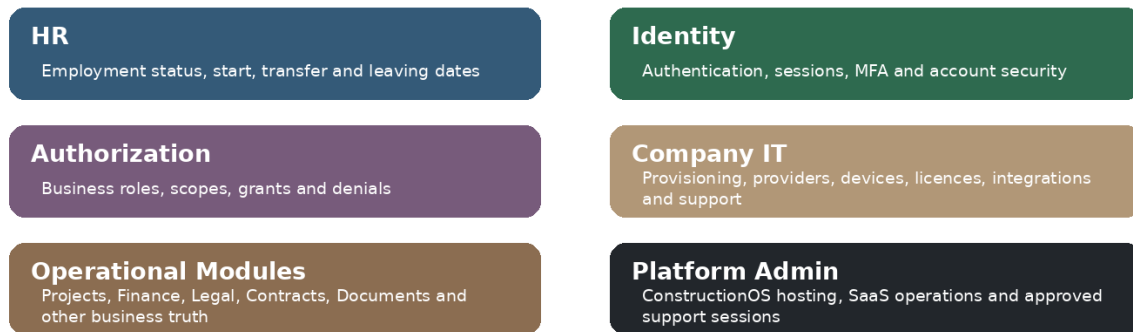
The module is mandatory as a platform capability but does not require a company to employ a dedicated IT team. Small companies may assign one person multiple IT roles, while enterprise groups may operate shared-service IT across selected companies.

The central product risk is authority confusion. An IT administrator may configure an organization-wide provider but must not automatically receive access to email content, private calendars, Finance, Payroll, Legal, Contracts, Documents or project records. Every access path remains subject to Authorization and the owning business domain.

Implementation decision

Provision a protected IT Department for every company and implement IT as a first-class bounded context with provider-neutral adapters, secrets isolation, service management records and explicit source-module boundaries.

IT Administration - Source-of-Truth Boundary



Company IT technically enables access. It does not automatically gain business-data authority.

Figure 1. Company IT technically enables the platform while business domains retain authority.

2. Product Vision and Business Outcomes

2.1 Product vision

ConstructionOS should allow each company to securely connect the systems it already uses while retaining one controlled operating model. Employees should receive accounts, licences, devices and support through traceable workflows. IT teams should understand every provider connection, permission scope, synchronization failure and critical dependency without relying on personal spreadsheets or undocumented administrator accounts.

2.2 Business outcomes

- Create one company-level inventory of identities, providers, applications, devices, licences and technical services.
- Reduce onboarding and offboarding delays through coordinated HR-triggered provisioning and revocation.
- Enable Microsoft 365, Outlook and Google Workspace without provider-specific coupling in operational modules.

- Prevent shadow integrations and excessive provider permissions.
- Reduce licence waste and recover entitlements during role change or offboarding.
- Provide an IT Service Desk for employee requests, incidents, changes and knowledge.
- Make integration failures, token expiry, device risk and access-review obligations visible before they become outages.
- Preserve business-data confidentiality while allowing IT to operate technical systems.

2.3 Failure risks to prevent

Risk	Required prevention
IT becomes a super-admin for business data	Separate technical capabilities from business-domain permissions.
Provider connection requests excessive scope	Plain-language scope review, least privilege and approval.
Offboarded user retains external access	Effective-dated revocation, session/device/provider enforcement and reconciliation.
Calendar sync overwrites governed records	Explicit authority, version mapping and conflict review.
Credentials leak through logs or tables	Dedicated secrets manager and redacted audit.
Support request becomes an untracked message	Structured service records, SLA and ownership.
Physical and technical device records diverge	Assets owns the physical asset; IT owns technical profile and compliance.
Platform Admin enters tenant operations	Approved, scoped Support Session only.

3. Success Metrics and Guardrails

Outcome	Metric / target
Provisioning	95% of standard joiners technically ready by configured start-time SLA.
Offboarding	Critical sessions and provider access revoked within configured effective-time SLA.
Integration health	99% of active connections report current health state and owner.
Scope governance	100% of organization-wide scopes have approval evidence and review date.
Secrets	Zero plaintext secrets in application tables, logs or exports.
Licence efficiency	Unused and overallocated licences visible by company and product.
Service Desk	Response and resolution SLAs measured by priority and service.
Audit	Every privileged IT action contains actor, scope, reason, result and correlation ID.
Privacy	No IT role receives mailbox, payroll, legal or project content by default.
Resilience	Provider outage does not disable the ConstructionOS core.

4. Scope, Non-Goals and Assumptions

4.1 In scope

- Protected IT Department and company/group operating model
- Identity providers, domains, SSO, account mappings and provisioning
- Microsoft 365, Outlook, Teams and resource connections
- Google Workspace, Gmail delivery, Calendar and Meet connections

- Email provider and shared-mailbox configuration
- Connected Applications Registry and integration governance
- Provider adapters, synchronization, failure handling and reconciliation
- Credentials references, certificates, service accounts and secrets governance
- Devices, endpoint security profiles and mobile revocation
- Software products, licence pools, assignments and renewals
- IT Service Desk, service catalogue, incidents, problems, changes and knowledge
- Security operations, access reviews, project/site IT, audit and analytics

4.2 Non-goals

- Replacing Microsoft 365, Google Workspace or an enterprise identity provider
- Reading or archiving employee email content by default
- Replacing operational business modules
- Owning employment, project, Finance, Contracts or Documents records
- Providing full endpoint management without a future provider adapter
- Allowing company IT to operate ConstructionOS infrastructure
- Implementing vendor-specific behavior inside source modules
- Using service tickets as substitutes for Tasks, Workflow or source records

4.3 Assumptions

- A company may use local ConstructionOS authentication without an external provider.
- A person may have multiple company contexts and different provider accounts.
- Some integrations are user-delegated while others are organization-wide.
- Provider APIs, scopes, rate limits and capabilities may change.
- Every integration has a technical owner and a business/data owner.
- Company configuration controls which providers and features are enabled.

5. Terminology and Core Concepts

Term	Definition
IT Department	Protected company system department representing the technical governance function.
Connected Application	Registered external or internal application with owners, purpose, scopes, risks and lifecycle.
Integration Connection	Technical provider connection for one company and environment.
Provider Adapter	Replaceable implementation translating a provider API into ConstructionOS contracts.
Consent	Evidence that required provider permissions were approved.
Account Mapping	Relationship between a ConstructionOS identity and provider identity.
Resource Mapping	Relationship between an internal resource and provider resource.
Service Account	Non-human technical identity with narrow capabilities, owner and expiry/review.
Configuration Item	Technical component tracked in the IT configuration registry.
Service Request	User request for a standard IT service.
Incident	Unplanned interruption or degradation.
Problem	Investigation of recurring or material incident causes.
Change	Controlled modification to technical services or configuration.
JML	Joiner, mover and leaver technical lifecycle.

Term	Definition
Integration Health	Current connection, credential, scope, sync and failure state.

6. Architectural Principles

Principle	Implementation meaning
Technical authority is not business authority	IT may administer providers and accounts but does not inherit business-record access.
Provider neutrality	Source modules integrate through ConstructionOS contracts, not Microsoft- or Google-specific code.
Core independence	ConstructionOS remains operational when external providers are unavailable.
Least privilege	Connections request only necessary scopes and require review when scope expands.
Secrets isolation	Application databases store secret references, never raw secrets.
Source ownership	Each business domain remains authoritative for its records and consequences.
Commands and events	Cross-domain actions use APIs, commands, events and projections.
Idempotency and reconciliation	Synchronization tolerates duplicates and verifies source-target consistency.
Effective-dated control	Access, licences, mappings and approvals have start/end and review dates.
Complete traceability	Privileged IT operations are immutable-audited.

7. Domain Ownership and Source-of-Truth Boundaries

Capability	Authoritative owner	IT responsibility
Employment and effective dates	HR	Consume events and execute technical provisioning/revocation.
Authentication, MFA and sessions	Identity	Configure providers and technical policy.
Business roles and permissions	Authorization	Apply approved changes; do not independently invent authority.
Physical devices and custody	Assets	Maintain technical device profile and compliance.
Licences and technical subscriptions	IT	Manage allocation, usage and renewal readiness.
Purchasing	Procurement	Request acquisition and link PO.
Contracts and vendor terms	Contracts	Link service/application to contract and SLA.
Invoices and payments	Finance	Consume cost and payment-status projection only.
Calendar and meetings	Calendar	Configure provider mapping and sync.
Notification delivery	Notifications	Configure email provider and sending identities.
Files and revisions	Documents	Configure connector; reference Document IDs.

Capability	Authoritative owner	IT responsibility
Business approvals	Workflow	Provide hooks and execute approved technical actions.
Platform infrastructure	Platform Admin	Company IT receives tenant-safe status only.

8. Personas, Roles and Stakeholders

Persona	Primary responsibilities	Default restrictions
Company Owner	Final company authority, high-risk consent and IT governance	Cannot bypass immutable audit or source-domain rules.
IT Director	Strategy, risk, architecture and major investment	No automatic operational data access.
IT Manager	Team, services, providers, devices and licences	High-risk approvals may require Owner/Security.
System Administrator	Accounts, sessions, domains and settings	No Company Owner modification.
Identity & Access Admin	SSO, mappings, provisioning, reviews	Cannot approve own high-risk access.
Integration Administrator	Provider configuration, sync and failure resolution	Cannot expand scopes without approval.
Security Administrator	MFA, findings, privileged access and incidents	Confidential business access remains separate.
Device Administrator	Endpoint profile, compliance and revocation	Physical custody remains Assets.
Licence Manager	Pools, allocations, usage and renewals	Procurement/Finance remain authoritative.
Helpdesk Agent	Tickets, ordinary recovery and escalation	No high-risk role grants or provider consent.
IT Auditor	Read-only evidence and configuration review	No operational edits.
Employee	Self-service requests and knowledge	Own requests and permitted resources only.
Project/Site IT	Project-scoped devices and connectivity	No access outside assigned projects.

9. Information Architecture and Navigation

Navigation area	Primary content
Dashboard	Health, provisioning, devices, licences, tickets, security and renewals.
People & Access	Provisioning, mappings, reviews, service accounts and JML.
Identity Providers	SSO, domains, directory synchronization and MFA.
Microsoft 365	Tenant, users, Outlook, Teams, rooms and health.
Google Workspace	Organization, users, Calendar, Gmail, Meet and health.
Email & Calendar	Sending identities, shared mailboxes and calendar mappings.
Connected Applications	Registry, consent, scopes, owners, mappings and lifecycle.
Integration Monitoring	Jobs, failures, rate limits, conflicts and reconciliation.
Devices	Technical profiles, compliance, revocation and software status.
Software & Licences	Products, pools, assignments, usage and renewals.
Service Desk	Requests, incidents, problems, changes, knowledge and SLA.

Navigation area	Primary content
Security Operations	Findings, privileged activity, access reviews and break-glass.
CMDB	Configuration items, services and dependencies.
Audit & Reports	Immutable actions, operational metrics and compliance evidence.

10. Company Provisioning and IT Organisation

Every company receives a protected IT Department at creation. The platform may hide or merge default teams for a small-company experience, but the capability structure remains intact.

Provisioned object	Rule
IT Department	System-created, renameable, not permanently deletable.
IT Administration team	Default technical administration team.
Identity & Access team	Default identity and access-management team.
Integrations team	Default provider and connected-application team.
IT Support team	Default Service Desk team.
Security Operations team	Default security and privileged-access team.
Devices & Software team	Default endpoint and licence team.
Initial authority	Assigned only by Company Owner or authorized governance workflow.
Group IT	Optional shared-service structure with explicit child-company scopes.

ID	Area	Requirement	Priority	Acceptance signal
IT-ORG-001	Provisioning	Every company shall receive one protected IT Department and baseline teams.	Must	Company creation transaction records generated IDs and audit.
IT-ORG-002	Governance	No user shall be assigned to IT automatically.	Must	New company has empty IT membership until Owner assignment.
IT-ORG-003	Deletion	The protected IT Department cannot be permanently deleted.	Must	Delete action is rejected; rename/hide remains available.
IT-ORG-004	Group	Group IT assignments must carry explicit company scopes.	Must	No implicit child-company access is observed.

11. IT Dashboard

Widget	Required information
Provisioning	Starting, failed, overdue and awaiting approval.
Offboarding	Scheduled revocations, overdue transfers and unrecovered assets.
Identity health	SSO state, domain verification, MFA and account conflicts.
Integrations	Active, degraded, suspended, expiring and failed.

Widget	Required information
Devices	Compliant, non-compliant, lost, revoked and unsupported.
Licences	Used, available, overallocated, unused and expiring.
Service Desk	Open by priority, SLA risk, aging and assignment.
Security	Critical findings, privileged changes, break-glass and suspicious sessions.
Changes	Upcoming windows, failed changes and rollback status.
Backup/continuity	Tenant-safe platform status and company-managed connector status.

ID	Area	Requirement	Priority	Acceptance signal
IT-DASH-001	Scope	Dashboard shall be company scoped and permission filtered.	Must	Sibling-company counts and details are absent.
IT-DASH-002	Freshness	Every health widget shows last verified time and source.	Must	Stale data is visibly labelled.
IT-DASH-003	Drill-down	Every actionable metric links to governed source records.	Must	No duplicate editable dashboard records exist.

12. IT Service Catalogue

Category	Example catalogue items
User & Access	New account, role request, MFA help, guest access, shared mailbox.
Devices	Laptop, mobile, monitor, repair, replacement, lost device, return.
Applications	Microsoft licence, Google licence, Autodesk, Adobe, BIM, new app.
Infrastructure	Office internet, printer, meeting room, site network, VPN.
Security	Phishing report, suspicious login, device compromise, exception.
Data & Recovery	Restore request, export request, backup validation.
Integration	New connection, mapping issue, provider failure, API access.

ID	Area	Requirement	Priority	Acceptance signal
IT-CAT-001	Configuration	Catalogue items shall be versioned and company configurable.	Must	Request stores exact catalogue-item version.
IT-CAT-002	Ownership	Each item shall define fulfilment team, SLA and approval route.	Must	Submitted request resolves all three.
IT-CAT-003	Visibility	Catalogue availability shall follow role, company and project scope.	Must	Unauthorized items are not discoverable.

13. Identity Provider Architecture

ConstructionOS shall expose one provider-neutral identity integration contract. Local authentication remains supported, while enterprise customers may connect Microsoft Entra, Google Workspace, SAML or OpenID Connect providers.

Connection field	Requirement
Provider identity	Provider type, provider tenant and environment.
Company ownership	Tenant and owning company.
Protocols	OIDC, SAML or supported provisioning protocol.
Domains	Verified domains and mapping policy.
Authentication mode	Local, SSO only, hybrid or emergency fallback.
Provisioning mode	Manual, import, sync, JIT or API.
Consent	Approved scopes, evidence, expiry and reviewer.
Credential	Secrets-manager reference and rotation state.
Health	Last test, token expiry, provider status and errors.
Version	Optimistic record version and configuration revision.

ID	Area	Requirement	Priority	Acceptance signal
IT-IDP-001	Provider model	Source applications shall call a provider-neutral identity contract.	Must	No source module contains provider-specific authentication logic.
IT-IDP-002	Fallback	Provider failure shall follow configured fallback without weakening policy.	Must	Tests prove correct fail-closed/fallback behavior.
IT-IDP-003	Domains	Domain ownership shall be verified before automatic mapping.	Must	Unverified domain cannot activate automatic mapping.
IT-IDP-004	Health	Connection health shall include credential, consent and provider state.	Must	Dashboard distinguishes each failure class.

14. Domains, SSO and Authentication Policy

- Verify company domains and detect conflicts across related companies.
- Configure permitted authentication modes by company and user class.
- Require MFA for privileged IT roles.
- Support SSO test mode before mandatory enforcement.
- Preserve local emergency access only under configured security policy.
- Record policy version used for every authentication decision.

Policy area	Required controls
Domain	Verification evidence, status, owner, aliases and conflict review.
SSO enforcement	Pilot population, activation time, exclusions and fallback.
MFA	Required user classes, methods and grace handling.
Session	Timeout, device/session list, revocation and risk response.
Emergency access	Named accounts, strong controls, alerts and reviews.
Company transfer	Re-evaluate provider account, domain and access.

15. User Provisioning and Account Mapping

Provisioning mode	Behavior
Manual	Authorized IT user creates or links one account.
Invite	User receives controlled activation invitation.
HR-triggered	HR event creates a provisioning case; IT/Authorization execute.
Directory import	Candidates staged, validated and previewed before commit.
Directory synchronization	Incremental mappings and lifecycle rules.
Just-in-time	Account created after successful provider login under policy.
API	Versioned, idempotent provisioning command.

Mapping outcome	Meaning
Matched automatically	Unique policy-compliant match.
Matched manually	IT selected verified target.
Unmatched	No candidate found.
Duplicate candidate	More than one plausible user.
Externally suspended	Provider account unavailable.
Externally removed	Provider object no longer exists.
Conflict	Mapping violates company, domain or identity rule.

ID	Area	Requirement	Priority	Acceptance signal
IT-USR-001	Boundary	Provisioning shall not silently grant business permissions.	Must	Account exists without unapproved capabilities.
IT-USR-002	Matching	Ambiguous identities shall never auto-merge.	Must	Duplicate candidates enter review.
IT-USR-003	Idempotency	Repeated provisioning commands shall return the same account result.	Must	No duplicate account is created.
IT-USR-004	Scope	Every mapping shall validate tenant and company ownership.	Must	Cross-company mapping is rejected.

16. Joiner, Mover and Leaver

Joiner - Mover - Leaver Technical Control

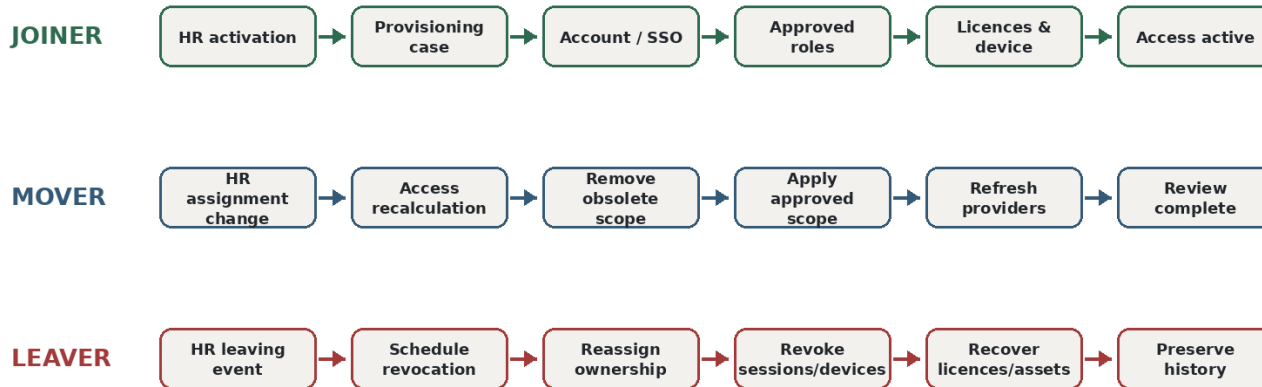


Figure 2. HR triggers effective-dated technical control while source histories remain independent.

16.1 Joiner

1. Receive HR start event or approved manual request.
2. Create provisioning case and resolve company identity.
3. Create or link account.
4. Apply approved authorization assignments.
5. Assign licences and request device/asset.
6. Configure provider and calendar mappings.
7. Verify MFA and device policy.
8. Activate at configured effective time.

16.2 Mover

9. Capture effective company, department, position or project change.
10. Calculate obsolete and required access.
11. Route additional authority for approval.
12. Revoke obsolete scopes and provider resources.
13. Apply approved scopes, licences and mappings.
14. Refresh mobile, search and integration projections.
15. Preserve previous employment and authorization history.

16.3 Leaver

16. Receive effective leaving event.
17. Identify sessions, devices, mappings, licences, service accounts and owned work.
18. Require ownership transfer or exception for active tasks and approvals.
19. Revoke sessions, provider access and mobile workspaces at effective time.
20. Recover licences and coordinate asset return.
21. Preserve records, audit and contribution history.
22. Reconcile provider and ConstructionOS states.

ID	Area	Requirement	Priority	Acceptance signal
IT-JML-001	Effective time	Revocation shall execute at the authoritative effective timestamp.	Must	Scheduled action and actual result are independently audited.

ID	Area	Requirement	Priority	Acceptance signal
IT-JML-002	Ownership	Open tasks, approvals and service identities require transfer or exception.	Must	Readiness view lists unresolved ownership.
IT-JML-003	History	Offboarding shall never delete historical records or contribution.	Must	Links remain valid after account suspension.
IT-JML-004	Reconciliation	Provider state shall be reconciled after every critical JML action.	Must	Mismatch creates a review item.

17. Microsoft 365 and Outlook

17.1 Connection profiles

- Organization-wide Microsoft tenant
- Selected-user delegated connection
- Application-based background connection
- Shared mailbox
- Room/resource mailbox
- Pilot/test environment

17.2 Supported capabilities

Capability	Initial behavior
Directory mapping	Read selected profile attributes and map accounts.
Provisioning	Create/disable provider accounts where approved and supported.
Outlook Calendar	One-way, two-way or free/busy-only synchronization.
Teams meetings	Generate meeting links from Calendar workflows.
Rooms/resources	Map provider resource mailboxes to Calendar resources.
Shared mailboxes	Map approved addresses and membership.
Notification email	Send through approved company sender.
Contacts	Optional selected-contact synchronization.
Files	SharePoint/OneDrive connector remains controlled roadmap unless enabled.

17.3 Authority and conflict

Each mapping must declare whether the internal or external record is authoritative. ConstructionOS-owned meetings, Outlook-owned personal events and source-module calendar projections must remain distinguishable. Controlled meeting decisions and final minutes can never be overwritten by Outlook synchronization.

ID	Area	Requirement	Priority	Acceptance signal
IT-MS-001	Consent	Organization-wide Microsoft access requires approved scope evidence.	Must	Activation blocked until approval exists.
IT-MS-002	Calendar	Every external event mapping stores internal/external versions and authority.	Must	Conflict can be reproduced and explained.

ID	Area	Requirement	Priority	Acceptance signal
IT-MS-003	Privacy	Free/busy sync shall not expose private titles or participants.	Must	Unauthorized availability query returns redacted interval only.
IT-MS-004	Resilience	Microsoft outage shall not prevent internal Calendar operation.	Must	Internal meeting creation remains available.

18. Google Workspace

Google Workspace shall use the same canonical account, calendar, email, resource, consent and synchronization contracts as Microsoft 365. Provider-specific fields remain inside the adapter configuration.

Google capability	ConstructionOS behavior
Organization connection	Company-owned provider connection and verified domains.
Directory mapping	Map provider identities to ConstructionOS users.
Google Calendar	Configured direction, privacy and conflict policy.
Gmail delivery	Approved sending identity for Notifications.
Google Meet	Generate meeting links from Calendar.
Calendar resources	Map rooms and resources.
Contacts	Optional selected-contact synchronization.
Drive	Future controlled connector unless separately enabled.

ID	Area	Requirement	Priority	Acceptance signal
IT-GOOG-001	Contract	Google adapter shall implement the shared provider interface.	Must	Calendar and identity modules call generic contracts.
IT-GOOG-002	Mapping	Provider resource IDs shall never become internal primary IDs.	Must	Stable internal mapping survives provider renames.
IT-GOOG-003	Revocation	External consent revocation shall suspend affected jobs immediately.	Must	Health state and owners are notified.

19. Email Delivery and Shared Mailboxes

IT configures email provider, domains, sender identities and shared mailboxes. Notifications & Event Centre remains responsible for recipient resolution, templates, policy, delivery attempts and user communication history.

Configuration	Required fields / controls
Sending domain	Domain, verification, DKIM/SPF status where available and owner.
Sender identity	Address, display name, purpose, allowed modules and status.
Reply-to	Default and workflow-specific policy.
Shared mailbox	Provider ID, company, owners, members and use cases.
Rate policy	Provider limits and company throttles.

Configuration	Required fields / controls
Test	Delivery address, result, provider reference and timestamp.
Failure	Bounce, rejection, authentication, rate limit and provider outage.
Branding	Company name, footer, legal text and classification policy.

ID	Area	Requirement	Priority	Acceptance signal
IT-MAIL-001	Boundary	Operational modules shall not send email directly.	Must	All delivery requests pass through Notifications.
IT-MAIL-002	Sender	Only approved sender identities may be selected.	Must	Unapproved address is rejected by backend.
IT-MAIL-003	Mailbox access	Managing a shared mailbox mapping does not grant message-content access.	Must	IT API responses contain configuration only.
IT-MAIL-004	Secrets	SMTP/provider credentials shall be stored by secret reference.	Must	No credential appears in DB export or audit.

20. Calendar, Meetings and Resource Integration

Owned by IT	Owned by Calendar & Meetings
Provider connection	Meeting identity and lifecycle
External calendar mapping	Agenda and participants
Room/resource mailbox mapping	Availability and reservation
Teams/Meet provider configuration	Attendance, decisions and minutes
Synchronization health	Action-item and Task mapping
Provider credentials and consent	Internal permission-safe projection

ID	Area	Requirement	Priority	Acceptance signal
IT-CAL-001	Source ownership	Rescheduling a source-owned date shall route to the source module.	Must	Calendar/IT cannot directly rewrite source record.
IT-CAL-002	Resource	Exclusive resource mappings shall preserve transactional reservation rules.	Must	No provider sync creates double booking.
IT-CAL-003	Privacy	IT may inspect mapping metadata but not private event content by default.	Must	Field-level API tests pass.
IT-CAL-004	Conflict	Controlled conflicts shall create review items, never silent last-write-wins.	Must	Conflict record contains versions and authority.

21. Connected Applications Registry

Every external system and high-impact internal technical application must be registered before production activation.

Field group	Required data
Identity	Name, provider, category, environment and status.
Ownership	Technical owner, business owner, data owner and support owner.
Scope	Companies, users, modules, projects and data classes.
Commercial	Supplier, contract, PO, subscription and renewal.
Technical	Connection type, adapter, endpoints, authentication and health.
Risk	Criticality, data classification, residency and exit strategy.
Governance	Consent, approval, review date and lifecycle.
Operations	SLA, incidents, changes, failures and runbook.
Audit	History of scopes, credentials, owners and activation.

ID	Area	Requirement	Priority	Acceptance signal
IT-APP-001	Registration	Production connection shall require a registered Connected Application.	Must	Unregistered activation command is rejected.
IT-APP-002	Ownership	Every application shall have technical, business and data owners.	Must	Activation blocked when owner missing.
IT-APP-003	Review	High-risk applications shall require periodic review.	Must	Overdue review creates governance finding.
IT-APP-004	Exit	Critical applications shall record revocation and exit plan.	Should	Plan visible before activation.

22. Integration Request, Review and Lifecycle

Connected Application Lifecycle

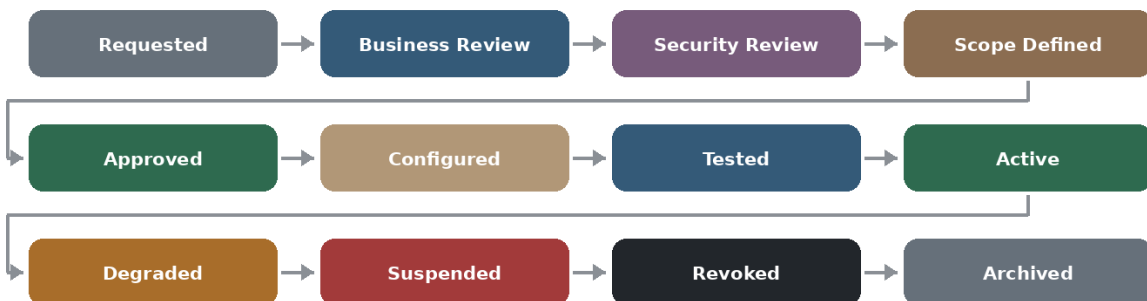


Figure 3. Integration governance spans business need, risk, configuration, health and retirement.

22.1 Integration request

- Business purpose and expected outcome
- Requester and affected companies
- Provider and environment
- Required users, modules and data
- Read/write actions and automation
- Expected volume and criticality
- Cost and commercial references
- Target activation date

22.2 Required review

23. Business owner confirms necessity.
24. Data owner confirms data use.
25. Security reviews scopes, authentication and exposure.
26. Legal/compliance reviews sensitive or cross-border processing where configured.
27. IT confirms technical support and exit plan.
28. Workflow records approval before activation.

ID	Area	Requirement	Priority	Acceptance signal
IT-LIFE-001	Activation	Connection activation requires approved request, test and owner confirmation.	Must	State cannot skip mandatory gates.
IT-LIFE-002	Scope change	Material permission expansion shall return to review.	Must	Old consent remains historical.
IT-LIFE-003	Suspension	Suspension stops new operations but preserves mappings and audit.	Must	Jobs remain paused and resumable.
IT-LIFE-004	Revocation	Revocation disables credentials, jobs and future access.	Must	Pending work receives deterministic result.

23. Integration Scopes, Consent and Data Classification

Scope class	Examples
Directory metadata	Name, email, provider ID and account status.
Calendar metadata	External ID, time, busy state and selected fields.
Calendar content	Title, participants, description and attachments.
Email delivery	Send as approved company identity.
Mailbox content	Highly restricted; not included by default.
File access	Selected libraries, folders or documents.
Administrative	User provisioning, suspension or provider resource management.
Background access	Application acts without interactive user session.

Data class	Default connector posture
Public	May be processed under approved purpose.
External Shared	Allowed for defined external workflows.
Internal	Requires company-approved connection.

Data class	Default connector posture
Commercial Restricted	Narrow scopes and explicit business owner.
Financial Restricted	Separate approval and field restrictions.
HR Confidential	No access without HR/data-owner policy.
Payroll Restricted	Excluded by default.
Legal Restricted	Matter-level and privilege review.
HSE Sensitive	Personal/medical details excluded by default.
Executive Restricted	Explicit named scope only.

ID	Area	Requirement	Priority	Acceptance signal
IT-SCP-001	Least privilege	Requested scopes shall be the minimum required for the approved purpose.	Must	Review UI compares requested and approved scopes.
IT-SCP-002	Expansion	New scope shall not activate silently after provider re-consent.	Must	Connection enters review/suspended state.
IT-SCP-003	Evidence	Consent shall record approver, scopes, time and provider reference.	Must	Evidence is immutable and queryable.
IT-SCP-004	Classification	Connection shall reject data classes outside its approved profile.	Must	Violation creates security finding.

24. Provider Adapter and Integration Runtime

The integration platform shall isolate provider-specific APIs behind versioned adapters. Operational modules must not call Microsoft, Google or other provider APIs directly.

Runtime component	Responsibility
Connection Management	Lifecycle, ownership, status and configuration.
Provider Adapter Registry	Provider type, supported capabilities and adapter versions.
Consent Service	Scopes, evidence and review.
Account Mapping	Internal/external identity mapping.
Resource Mapping	Calendar, mailbox, room and other mappings.
Synchronization Engine	Pull, push, webhook and polling jobs.
Token Coordinator	Secret references, refresh and expiry.
Webhook Gateway	Signature validation, rate control and routing.
Retry / Dead Letter	Recoverable failure and operator review.
Reconciliation	Source-target comparison and repair of projections.
Health Service	Provider, credential, scope, rate and sync status.

Adapter operation	Contract expectation
authorize	Start provider authorization and return state reference.

Adapter operation	Contract expectation
refreshCredentials	Refresh through secrets layer.
revoke	Revoke provider access and update local state.
testConnection	Return structured capability and health result.
listAccounts	Cursor-paginated provider identities.
listResources	Selected provider resources.
pushChange	Versioned outbound command.
pullChanges	Incremental cursor or time-window fetch.
processWebhook	Validated idempotent webhook processing.
getHealth	Provider, credential, rate and capability state.

25. Synchronization, Conflict and Reconciliation

Sync method	Use
Webhook	Near-real-time provider changes.
Delta cursor	Incremental changes without full scans.
Scheduled polling	Providers lacking reliable events.
Outbound command	ConstructionOS-initiated change.
On-demand refresh	User/operator verification.
Manual import	Controlled migration or exceptional use.

Conflict outcome	Rule
Internal wins	Only where internal source is authoritative.
External wins	Only where provider source is authoritative.
Merge	Only for explicitly mergeable fields.
User review	Human decision with before/after versions.
Source owner decision	Route to authoritative business module.
Suspend mapping	Stop unsafe synchronization until resolution.

ID	Area	Requirement	Priority	Acceptance signal
IT-SYNC-001	Idempotency	Repeated webhook/job input shall not duplicate effects.	Must	Unique provider object/version constraint passes.
IT-SYNC-002	Ordering	Mappings shall detect late and out-of-order provider changes.	Must	Older version cannot overwrite newer projection.
IT-SYNC-003	Conflict	Controlled records shall never use silent last-write-wins.	Must	Conflict record is generated.
IT-SYNC-004	Reconciliation	Critical mappings shall support scheduled and on-demand reconciliation.	Must	Missing/stale/orphan outcomes are reported.

ID	Area	Requirement	Priority	Acceptance signal
IT-SYNC-005	Partial failure	Batch processing shall report per-item success and failure.	Must	Safe items commit without hiding failed items.

26. Credentials, Secrets and Certificates

Stored in IT database	Stored in secrets manager
Secret reference	Access/refresh tokens
Credential type	Client secret
Provider and connection	Private key
Created/expiry/rotation times	Certificate private material
Scope and status	Passwords
Last successful use	Webhook signing secret

ID	Area	Requirement	Priority	Acceptance signal
IT-SEC-001	Storage	Plaintext credentials shall never be stored in ordinary tables.	Must	Automated secret scan passes.
IT-SEC-002	Logging	Secrets and token values shall be redacted from logs and audit.	Must	Security test finds no credential material.
IT-SEC-003	Rotation	Credentials shall support scheduled and emergency rotation.	Must	Rotation retains trace and validates new credential.
IT-SEC-004	Access	Secrets access shall be service-scoped and least privilege.	Must	Human UI cannot retrieve raw secret.

27. Service Accounts and Automation Identities

Field	Requirement
Identity	Unique non-human ID and name.
Owner	Named team and accountable person.
Purpose	Documented business/technical purpose.
Capabilities	Explicit narrow permissions.
Scope	Company, project, API and environment.
Credential	Secret reference and rotation.
Lifecycle	Effective date, expiry/review and status.
Monitoring	Last use, unusual activity and failures.
Restrictions	No UI login; no Owner/Admin role templates.

ID	Area	Requirement	Priority	Acceptance signal
IT-SA-001	Ownership	Every service account shall have an active accountable owner.	Must	Owner departure triggers reassignment finding.
IT-SA-002	Expiry	Service accounts shall have expiry or review date.	Must	Overdue account is flagged/restricted by policy.
IT-SA-003	Privilege	Service accounts cannot receive Company Owner or broad Company Admin.	Must	Assignment rejected.
IT-SA-004	Use	Every call shall identify service account and correlation context.	Must	Audit can trace automated action.

28. Devices and Endpoint Administration

Assets remains authoritative for the physical device, purchase, custody and disposal. IT owns technical profile, security posture, software state and remote technical actions.

IT device field	Examples
Identity	Asset ID, serial/device ID and device type.
Platform	Operating system and version.
Security	Encryption, screen lock, security agent and compliance.
Management	Managed/unmanaged and provider reference.
Assignment	User, company, project and effective period.
Connectivity	Last contact and app version.
Risk	Lost, stolen, compromised or unsupported.
Actions	Revoke, restrict, remote wipe request and retire.

ID	Area	Requirement	Priority	Acceptance signal
IT-DEV-001	No duplication	IT technical profile shall reference the Assets record.	Must	No duplicate physical device master exists.
IT-DEV-002	Revocation	Lost-device action shall revoke sessions and mobile workspace.	Must	Revocation propagation is observed.
IT-DEV-003	Remote wipe	Remote wipe shall require configured approval and evidence.	Must	Unapproved request cannot execute.
IT-DEV-004	Compliance	Device policy shall restrict sensitive actions where configured.	Must	Non-compliant device receives deterministic deny.

29. Software, Licences and Subscriptions

Entity	Purpose
Software Product	Canonical application identity and vendor.
Edition/Plan	Product tier and capability.

Entity	Purpose
Licence Pool	Company entitlement, quantity and terms.
Licence Assignment	User/device/project allocation and effective dates.
Subscription Renewal	Renewal date, owner, budget and workflow.
Usage Snapshot	Provider or internal usage state.
Compliance Finding	Overallocation, unused, expired or prohibited use.

ID	Area	Requirement	Priority	Acceptance signal
IT-LIC-001	Allocation	Assignment shall validate entitlement availability and user scope.	Must	Overallocation follows configured block/warn policy.
IT-LIC-002	Recovery	Offboarding and expiry shall create licence-recovery actions.	Must	Recovered entitlement returns to pool.
IT-LIC-003	Commercial	Licence links shall reference Procurement, Contracts and Finance records.	Must	No duplicate invoice or contract data.
IT-LIC-004	Renewal	Renewals shall trigger owner notification and workflow by threshold.	Must	Upcoming renewal appears on dashboard.

30. IT Asset Management

IT asset operations extend the Assets Module through technical profiles, service relationships and maintenance references. Physical handover, location and condition history remain in Assets.

IT asset class	Examples
Endpoint	Laptop, desktop, tablet and phone.
Network	Router, switch, access point and firewall.
Meeting room	Display, camera, microphone and controller.
Office equipment	Printer, scanner and label device.
Server/storage	Server, NAS and storage appliance.
Security	Access device, token and security appliance.
Site IT	Temporary router, project workstation and site printer.

ID	Area	Requirement	Priority	Acceptance signal
IT-AST-001	Handover	Technical provisioning shall not replace Assets handover and custody.	Must	Both linked records exist where required.
IT-AST-002	Maintenance	IT repair/update work shall link to Assets maintenance or Task.	Must	No duplicate maintenance truth.
IT-AST-003	Disposal	Technical wipe confirmation shall be required before disposal where configured.	Must	Asset disposal gate receives IT result.

31. Service Desk and Employee Support Portal

IT Service Management Record Relationships

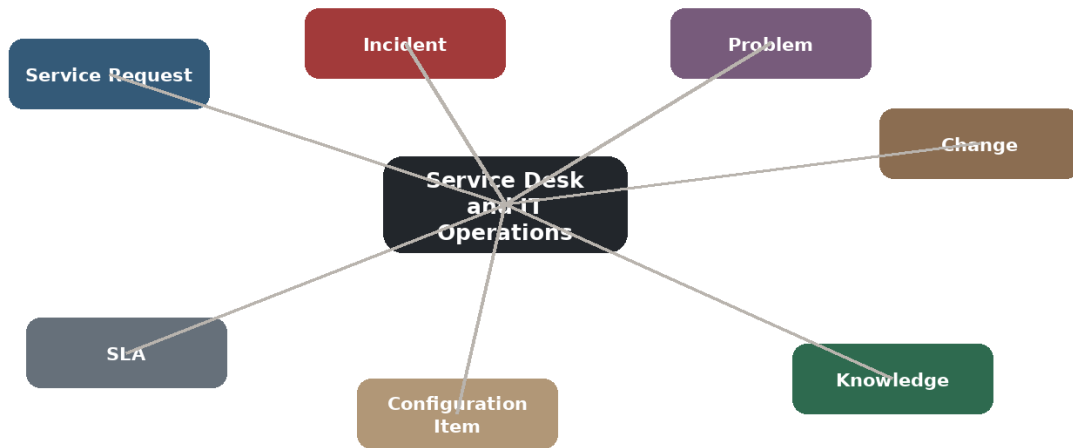


Figure 4. Service records remain linked but retain distinct purposes and lifecycles.

The Service Desk provides a governed intake and resolution environment. Tasks may be created for execution, but ticket lifecycle remains owned by IT Service Management.

Record	Purpose
Service Request	Standard user need or fulfilment request.
Incident	Restore a degraded or unavailable service.
Problem	Investigate underlying cause and prevention.
Change	Control technical modification and rollback.
Access Request	Obtain or change technical/business access through Authorization.
Security Incident	Restricted investigation and response.
Knowledge Article	Reusable approved guidance.
SLA	Response and resolution commitments.

32. Service Requests and Access Requests

Ticket field group	Required data
Identity	Ticket number, type, category and status.
People	Requester, affected user, manager and assigned team/agent.
Context	Company, department, project, site, device and application.
Priority	Impact, urgency, priority and SLA.
Request	Description, catalogue item, requested date and evidence.
Control	Approval, dependencies, risk and confidentiality.
Resolution	Outcome, user confirmation, linked task/change and closure.

Ticket field group	Required data
Audit	Timeline, comments, status changes and notifications.

ID	Area	Requirement	Priority	Acceptance signal
IT-REQ-001	Catalogue	Standard request shall use a versioned catalogue item.	Must	Request stores exact version and SLA.
IT-REQ-002	Access	Business-access request shall execute only through Authorization.	Must	IT ticket alone creates no permission.
IT-REQ-003	SLA	SLA timers shall pause only under configured waiting states.	Must	Timer history is explainable.
IT-REQ-004	Closure	Closure shall preserve resolution, evidence and requester response.	Must	Closed ticket remains auditable.

33. Incident and Major Incident Management

Incident attribute	Examples
Impact	Single user, department, project, company or group.
Urgency	Workaround, deadline, safety or financial dependency.
Service	Identity, email, calendar, application, device or network.
Priority	Configured impact x urgency matrix.
Timeline	Detected, acknowledged, contained, restored and closed.
Communication	Stakeholder updates through Notifications.
Evidence	Logs, screenshots, provider references and linked documents.
Outcome	Restoration, workaround, linked problem and review.

33.1 Major Incident roles

- Incident Commander
- Technical Lead
- Security Lead
- Business Owner
- Communications Owner
- Vendor Coordinator
- Scribe / Timeline owner

ID	Area	Requirement	Priority	Acceptance signal
IT-INC-001	Priority	Priority shall be calculated from configurable impact and urgency.	Must	Matrix result is visible and auditable.
IT-INC-002	Major incident	Critical incident shall support commander, update cadence and review.	Must	Required roles and updates are tracked.
IT-INC-003	Communication	Incident records shall publish communication intent, not send directly.	Must	Notifications owns delivery.

ID	Area	Requirement	Priority	Acceptance signal
IT-INC-004	Problem link	Recurring/significant incident may create one linked Problem.	Should	Duplicate root-cause investigations are warned.

34. Problem and Known Error Management

Problem area	Required data
Symptoms	Observed failures and affected services.
Incidents	Linked incidents and frequency.
Analysis	Hypotheses, evidence and root cause.
Known error	Documented cause and workaround.
Fix	Linked change, release or vendor action.
Prevention	Corrective actions and monitoring.
Closure	Validation period and outcome.

ID	Area	Requirement	Priority	Acceptance signal
IT-PRB-001	Root cause	Problem closure shall require cause or documented accepted unknown.	Must	Closure reason is structured.
IT-PRB-002	Known error	Published workaround shall link to approved Knowledge Article.	Should	Helpdesk can discover current workaround.
IT-PRB-003	Change	Corrective technical change shall use Change Management.	Must	Problem cannot directly alter production configuration.

35. IT Change Management

Change type	Governance
Standard	Pre-authorized template with repeatable low-risk steps.
Normal	Assessment, approval, scheduling, test and rollback.
Emergency	Expedited execution plus mandatory post-event review.
Integration	Provider, scope, mapping or synchronization change.
Security	MFA, identity, credential or policy change.
Migration	Data/configuration move with reconciliation.
Deployment	Application or software release.
Infrastructure	Network, device, service or configuration change.

Required plan	Content
Business reason	Need and expected outcome.

Required plan	Content
Risk assessment	Impact, likelihood and affected services.
Implementation	Ordered execution steps and owner.
Testing	Pre- and post-change checks.
Rollback	Safe reversal criteria and steps.
Window	Start/end, blackout and communication.
Approval	Required decision roles and evidence.
Result	Completed, failed, rolled back or partially completed.

ID	Area	Requirement	Priority	Acceptance signal
IT-CHG-001	Control	Production technical changes shall reference an approved Change when required.	Must	Direct high-risk configuration update is blocked.
IT-CHG-002	Rollback	Normal/high-risk changes shall include rollback plan.	Must	Approval blocked when plan missing.
IT-CHG-003	Emergency	Emergency change requires post-event review.	Must	Unreviewed emergency change appears as finding.
IT-CHG-004	SoD	Change implementer and final reviewer shall be separable.	Must	Configured SoD is enforced.

36. Configuration Management and CMDB

Configuration item type	Examples
Application	Microsoft 365, Google Workspace, BIM or internal service.
Integration	Provider connection or interface.
Identity provider	SSO/directory connection.
Infrastructure	Server, database, network or storage service.
Endpoint	Managed device or appliance.
Service account	Automated technical identity.
Domain/certificate	DNS domain or certificate.
Backup job	Company-managed external backup configuration.
Site service	Project/site connectivity and equipment.

Relationship	Meaning
depends_on	Service requires another item.
hosted_on	Application runs on platform/infrastructure.
connects_to	Technical connection exists.
authenticates_through	Identity provider dependency.
stores_data_in	Storage dependency.

Relationship	Meaning
used_by	Company/team/project usage.
supported_by	Responsible team/vendor.
backed_up_by	Recovery dependency.

ID	Area	Requirement	Priority	Acceptance signal
IT-CMDB-001	Boundary	CMDB shall not duplicate business records or source data.	Must	References link to authoritative domains.
IT-CMDB-002	Impact	Change/incident impact analysis shall use configuration relationships.	Should	Affected services are displayed.
IT-CMDB-003	Quality	Unowned, stale and orphaned items shall create data-quality issues.	Must	Quality dashboard lists findings.

37. Security Operations

Security area	Required capabilities
Identity	MFA compliance, privileged roles, sessions and risky accounts.
Devices	Non-compliance, lost/stolen and unsupported versions.
Integrations	Scope changes, token expiry, consent and suspicious failures.
Service accounts	Expiry, ownership, privilege and unusual use.
Exports	Restricted exports and sensitive downloads.
Break-glass	Activation, alerts, expiry and review.
Findings	Severity, containment, owner, evidence and closure.
Incidents	Restricted security investigation and communication.

ID	Area	Requirement	Priority	Acceptance signal
IT-SOC-001	Finding	Security findings shall have severity, owner and lifecycle.	Must	Critical unassigned finding triggers escalation.
IT-SOC-002	Confidentiality	Security incident details shall use independent restricted access.	Must	General IT staff cannot read restricted evidence.
IT-SOC-003	Break-glass	Emergency privileged access shall be time-limited and reviewed.	Must	Automatic expiry and post-review are enforced.
IT-SOC-004	No audit deletion	IT roles shall not modify or permanently delete audit.	Must	Backend rejects alteration.

38. Access Reviews and Privileged Access

Access reviews certify that current access remains appropriate. Reviews may target IT roles, Company Admins, external-provider administrators, Finance/Payroll roles, service accounts, shared mailboxes, project access or temporary grants.

Review step	Behavior
Campaign definition	Scope, reviewer, due date and policy version.
Population generation	Snapshot users, roles, resources and risk.
Review	Retain, modify, revoke, expire or escalate.
Execution	Authorization/IT execute approved changes.
Exception	Reason, owner and expiry.
Certification	Completion summary and evidence.
Follow-up	Failed revocation or unresolved conflict.

ID	Area	Requirement	Priority	Acceptance signal
IT-REV-001	Snapshot	Review shall use an immutable access snapshot.	Must	Later role changes do not alter reviewed evidence.
IT-REV-002	Execution	Review decision shall not directly mutate source access without command.	Must	Authorization confirms result.
IT-REV-003	Overdue	Critical privileged reviews shall escalate when overdue.	Must	Escalation event is published.

39. Project and Site IT

Project/site capability	Examples
Connectivity	Internet, router, access point and temporary network.
Devices	Site office computers, tablets, printers and scanners.
BIM	High-performance workstations and viewer devices.
Meeting	Site meeting-room equipment.
Access	Project-scoped accounts and guest access.
Software	Project licences and temporary subscriptions.
Closure	Asset return, licence recovery and data handover.

ID	Area	Requirement	Priority	Acceptance signal
IT-SITE-001	Scope	Project IT role shall be explicitly project/location scoped.	Must	No access to unrelated company projects.
IT-SITE-002	Assets	Site devices shall retain Assets custody and location history.	Must	Technical profile references Asset ID.
IT-SITE-003	Closure	Project closeout shall reconcile devices, licences and accounts.	Must	Unresolved resources block IT closeout readiness.

40. Knowledge Base and Self-Service

Article field	Requirement
Audience	IT-only, all employees, project, portal or helpdesk.
Language	English, Albanian and future locales.
Ownership	Author, reviewer and service owner.
Lifecycle	Draft, review, published, superseded and archived.
Version	Immutable published revision and effective dates.
Links	Service, application, error, ticket and configuration item.
Feedback	Helpful/not helpful and suggested update.
Expiry	Review date and stale-content warning.

ID	Area	Requirement	Priority	Acceptance signal
IT-KB-001	Publication	Only approved revision shall be visible to target audience.	Must	Draft is never exposed.
IT-KB-002	Access	Sharing an article shall not expand access to linked records.	Must	Linked resources recheck permissions.
IT-KB-003	Staleness	Expired articles shall be hidden or warned by policy.	Should	Owner receives review event.

41. Vendor, Contract, Procurement and Finance Integration

IT owns	Source domain owns
Technical service and application record	Supplier identity - Procurement
Licence usage and assignment	Purchase lifecycle - Procurement
Renewal requirement	Legal agreement - Contracts
Technical SLA performance	Invoice, payment and accounting - Finance
Provider health and incidents	Budget and financial approval - Finance
Warranty/service status	Official documents - Documents

ID	Area	Requirement	Priority	Acceptance signal
IT-COM-001	References	IT commercial views shall use source IDs and permission-safe projections.	Must	No duplicate contract/invoice records.
IT-COM-002	Renewal	Renewal request shall link current contract, supplier and cost owner.	Must	Workflow receives complete source references.
IT-COM-003	Visibility	IT cost visibility shall follow Finance field permissions.	Must	Unauthorized users receive no amounts.

42. Workflow and Approval Integration

Decision point	Typical approvers
Organization-wide provider connection	Company Owner/Admin, IT, Security and data owner.
Scope expansion	Business/data owner and Security.
SSO mandatory enforcement	Owner/Admin, IT and Security.
Automated provisioning/deprovisioning	HR, Authorization, IT and Security.
High-privilege service account	Service owner, Security and Authorization.
Remote wipe	IT/Security and configured company authority.
Emergency change	Emergency approver plus post-review.
Licence purchase/renewal	Business owner, Procurement and Finance.
Break-glass	Security authority and mandatory review.

ID	Area	Requirement	Priority	Acceptance signal
IT-WF-001	Exact version	Workflow submission shall reference exact IT record version.	Must	Material change invalidates/restarts decision.
IT-WF-002	Finalization	Approved decision shall be finalized by IT source service.	Must	Workflow completion alone does not change provider.
IT-WF-003	SoD	Technical configurator shall be separable from approver.	Must	Configured conflict is enforced.
IT-WF-004	Evidence	High-risk decision shall preserve scope and test snapshot.	Must	Audit can reconstruct approved state.

43. Notifications and Event Centre Integration

Event	Typical communication
Integration degraded	Technical owner and support team.
Consent/token expiring	Integration Admin and business owner.
Provisioning failed	IT owner, HR owner and requester.
Offboarding incomplete	HR, IT and manager.
Device non-compliant	User and Device Admin.
Licence expiring/overallocated	Licence owner and cost owner.
Ticket SLA risk	Assigned agent and manager.
Major incident	Configured stakeholders and update cadence.
Critical security finding	Security, Owner/Admin and required responders.
Access review overdue	Reviewer and escalation chain.

ID	Area	Requirement	Priority	Acceptance signal
IT-NOT-001	Boundary	IT shall publish facts/intents and not implement channel delivery.	Must	Notifications owns channel result.
IT-NOT-002	Minimization	Push/email payloads shall exclude secrets and restricted details.	Must	Security content test passes.
IT-NOT-003	Acknowledgement	Critical notices may require acknowledgement without implying approval.	Should	Acknowledgement state is separate.

44. Domain Data Model

Aggregate / entity	Purpose
ITDepartmentProfile	Company IT configuration and mode.
IdentityProviderConnection	SSO/directory provider configuration.
VerifiedDomain	Domain verification and mapping policy.
ExternalAccountMapping	Internal/provider identity relationship.
ProvisioningRequest	Durable account lifecycle command/case.
ConnectedApplication	Governed application registry record.
IntegrationConnection	Provider connection and lifecycle.
IntegrationScope / Consent	Approved permission profile and evidence.
IntegrationMapping	Account/resource/object mapping.
IntegrationSyncJob / Cursor	Synchronization execution and incremental state.
IntegrationConflict / Failure	Unsafe mismatch and operational failure.
IntegrationHealthSnapshot	Current connection health projection.
ITDeviceProfile	Technical extension to Assets device.
SoftwareProduct / LicencePool	Software and entitlement master.
LicenceAssignment	Effective-dated user/device allocation.
ITServiceRequest / Incident / Problem / Change	Service management aggregates.
ConfigurationItem / Relationship	CMDB technical dependency model.
SecurityFinding / SecurityIncident	Restricted security operations.
ServiceAccount	Non-human technical identity.
AccessReviewCampaign / Decision	Privilege certification.
KnowledgeArticle / Revision	Controlled support knowledge.

44.1 Canonical technical fields

Field	Rule
id	Immutable globally unique ID.

Field	Rule
tenant_id	Mandatory tenant boundary.
owning_company_id	Mandatory company owner for company records.
record_version	Optimistic concurrency version.
lifecycle_status	Controlled domain lifecycle.
confidentiality_class	Field/query policy input.
created/updated actor/time	Complete technical history.
correlation_id	Cross-module trace context.
source_reference	Typed authoritative source link where derived.
effective_from/to	Effective-dated assignments and grants.

45. Backend Services and Deployment Architecture

Service	Responsibility
IT Department Provisioning	Create protected company structures and defaults.
Identity Provider Service	Connections, domains and authentication configuration.
Provisioning Service	Durable JML/account actions.
Integration Registry	Connected applications, owners and lifecycle.
Consent & Scope Service	Approval evidence and permission profiles.
Provider Adapter Runtime	Provider-specific communication.
Synchronization Service	Jobs, webhooks, polling, cursors and mappings.
Credential Coordinator	Secrets references, expiry and rotation.
Reconciliation Service	Source-target consistency checks.
Device Administration	Technical profiles, compliance and actions.
Licence Service	Pools, assignments, usage and renewal.
Service Desk	Requests, SLA and fulfilment.
Incident/Problem/Change	ITSM controlled lifecycles.
CMDB Service	Configuration items and relationships.
Security Operations	Findings, restricted incidents and reviews.
Knowledge Service	Controlled knowledge revisions.
IT Reporting/Audit	Metrics, evidence and diagnostics.

45.1 Deployment decision

The initial release may operate within a modular platform deployment with a strongly separated IT schema and independent background workers. Provider adapter workers, synchronization queues, webhook processing and secret access should be separately scalable and fault-isolated. High-volume enterprise adapters may later become independent services without changing contracts.

46. API Contract

46.1 Standard request controls

- Authenticated tenant and explicit active company
- Capability and scope check
- Expected record version for controlled mutation
- Idempotency key for durable commands
- Correlation and trace IDs
- Structured validation errors
- Cursor pagination and filter schema
- Field-level response policy
- Reauthentication for high-risk actions

46.2 Representative endpoints

Area	Representative endpoints
Dashboard	GET /api/v1/it/dashboard; GET /api/v1/it/health
Identity	GET/POST /it/identity-providers; test, activate, suspend, revoke
Provisioning	GET/POST /it/provisioning-requests; execute, retry, cancel
Integrations	GET/POST/PATCH /it/integrations; test, activate, suspend, revoke, reconcile
Mappings	GET/POST /it/integrations/{id}/accounts resources objects
Failures	GET /it/integrations/{id}/failures; retry or dead-letter review
Devices	GET /it/devices; restrict, revoke, remote-wipe-request
Licences	GET /it/licences; assign, recover, renew-request
Tickets	GET/POST /it/tickets; assign, resolve, close, reopen
Incidents	GET/POST /it/incidents; declare-major, update, restore
Changes	GET/POST /it/changes; assess, approve, schedule, complete, rollback
Security	GET /it/security/findings; contain, resolve, review
Reviews	GET/POST /it/access-reviews; decide, certify

ID	Area	Requirement	Priority	Acceptance signal
IT-API-001	Authorization	Every endpoint shall enforce source-domain authorization.	Must	Gateway-only authorization is insufficient.
IT-API-002	Concurrency	Controlled update shall require expected version.	Must	Stale update returns conflict.
IT-API-003	Errors	Provider failures shall use stable internal error codes.	Must	UI never depends on raw provider message.
IT-API-004	Secrets	No API shall return raw credentials.	Must	Response-schema tests pass.
IT-API-005	Audit	Sensitive command shall record permission and actor context.	Must	Audit trace is complete.

47. Domain Events and Integration Events

Event family	Examples
Department	it.department.provisioned; it.role.assigned
Identity	it.identity_provider.connected; it.user.provisioned; it.user.deprovisioned
Integration	it.integration.approved; activated; degraded; revoked; sync_failed
Device	it.device.registered; compliance_changed; lost; revoked
Licence	it.licence.assigned; recovered; expiring; overallocated
Service Desk	it.ticket.submitted; assigned; escalated; resolved; sla_breached
Incident/Change	it.incident.major_declared; service_restored; it.change.failed
Security	it.security_finding.created; break_glass.activated; access_review.completed

47.1 Consumed events

- hr.employee.starting / transferred / leaving
- identity.user.created / suspended / session.revoked
- authorization.assignment.changed / permission.revoked
- assets.asset.assigned / returned / disposed
- contracts.contract.expiring
- procurement.purchase_order.issued
- finance.invoice.posted
- calendar.resource.created
- mobile.device.registered
- platform.company.read_only_started

ID	Area	Requirement	Priority	Acceptance signal
IT-EVT-001	Outbox	IT domain events shall be written through transactional outbox.	Must	Committed state cannot lose event.
IT-EVT-002	Consumer	Consumers shall be idempotent and version aware.	Must	Duplicate event produces one effect.
IT-EVT-003	Privacy	Events shall contain references/minimum data, not secrets or mailbox content.	Must	Schema review passes.
IT-EVT-004	Ordering	Entity events shall preserve aggregate version.	Must	Late event cannot regress projection.

48. Frontend and UX Architecture

Pattern	Use
IT list page	Integrations, devices, licences, tickets and applications.
Technical detail workspace	Connection, device, service account or application.
Health rail	Current state, last check, owner and next action.
Mapping table	Internal/external identity and resource relationships.
Scope diff	Requested, approved and provider-granted permissions.

Pattern	Use
Failure drawer	Structured error, attempts, safe retry and runbook.
Timeline	Lifecycle and operator actions.
Wizard	Provider connection, provisioning, request and change.
Conflict workspace	Version comparison, authority and resolution.
Sensitive-action modal	Reauthentication, approval and explicit confirmation.

48.1 UX state standards

- Empty and not configured
- Draft configuration
- Awaiting approval
- Testing
- Active and healthy
- Active but stale
- Degraded
- Authentication expired
- Scope mismatch
- Suspended
- Revoked
- Permission denied
- Version conflict
- Provider unavailable

ID	Area	Requirement	Priority	Acceptance signal
IT-UX-001	Clarity	Technical states shall use plain-language explanation and next action.	Must	User can distinguish configuration, provider and permission failures.
IT-UX-002	Safety	High-risk actions shall not use optimistic UI.	Must	Server confirmation precedes success state.
IT-UX-003	Privacy	Hidden restricted fields shall not be present in page payload.	Must	Network-response test passes.
IT-UX-004	Accessibility	Core IT workflows shall target WCAG 2.1 AA.	Must	Keyboard and screen-reader tests pass.

49. Roles, Capabilities and Permissions

Role	Typical capabilities	Critical restrictions
IT Director	IT governance, policy, risk and high-level approval	No automatic business-record access.
IT Manager	People, services, providers, devices and licences	Owner/Security gates remain.
System Administrator	Accounts, sessions, domains and technical settings	Cannot modify Company Owner.
Identity & Access Admin	SSO, mappings, provisioning and reviews	Cannot self-approve high-risk access.
Integration Admin	Connections, tests, mappings, sync and recovery	Cannot expand scopes without approval.
Security Admin	Findings, privileged access and break-glass review	Restricted incidents separately scoped.
Device Admin	Profiles, compliance and revocation	Remote wipe may require approval.
Licence Manager	Pools, allocations and renewals	No Finance posting.

Role	Typical capabilities	Critical restrictions
Helpdesk Agent	Tickets, ordinary recovery and escalation	No organization consent or high-risk grants.
IT Auditor	Read configuration and evidence	No edits.
Project IT	Assigned project/site support	No company-wide scope.

49.1 Core capability families

- it.department.*
- it.users.*
- it.identity_providers.*
- it.integrations.*
- it.integration_scopes.*
- it.devices.*
- it.licences.*
- it.service_desk.*
- it.incidents.*
- it.problems.*
- it.changes.*
- it.security.*
- it.access_reviews.*
- it.audit.view
- it.configure

Permission boundary

Membership in the IT Department does not grant every IT capability. IT roles and company/project scopes are evaluated independently, and business-domain permissions remain separate.

50. Security, Privacy and Compliance

Control	Requirement
Deny by default	No implicit access from department membership or job title.
Company isolation	Every command and mapping validates active company.
Field security	Mailbox, calendar, HR, Finance and Legal fields independently restricted.
Secrets	Dedicated manager; no plaintext in DB, logs or exports.
Provider scopes	Least privilege, consent and periodic review.
Privileged access	MFA, reauthentication, time limits and audit.
Webhooks	Signature validation, replay protection and rate control.
Data residency	Connections and derived data follow tenant policy.
Device policy	Sensitive actions may require trusted/managed device.
Break-glass	Named actor, reason, alert, expiry and post-review.
Support boundary	Platform Admin requires approved Support Session.
Testing	Tenant/company isolation and provider-security suite.

ID	Area	Requirement	Priority	Acceptance signal
IT-PRV-001	Mailbox	IT roles shall not receive email-content access by default.	Must	Mailbox-content endpoint is unavailable without explicit profile.
IT-PRV-002	Calendar	IT roles shall not receive private event detail by default.	Must	Only mapping/free-busy metadata is visible.
IT-PRV-003	Sensitive domains	HR/Payroll/Legal/Finance access remains independently authorized.	Must	Cross-domain tests return denied/filtered results.
IT-PRV-004	Support	Platform support cannot use company IT authority.	Must	Support requires active approved session.
IT-PRV-005	Deletion standby	Company read-only standby shall block IT mutations.	Must	All commands return governed read-only error.

51. Audit, Retention and Legal Hold

Audit category	Examples
Identity	Provider connected, domain verified, account mapped or deprovisioned.
Consent	Scope requested, approved, expanded or revoked.
Integration	Activated, tested, failed, retried, reconciled and revoked.
Credentials	Created, rotated, expired and revoked - reference only.
Devices	Registered, compliance changed, lost, restricted or wiped.
Licences	Assigned, recovered, overallocated and renewed.
Service management	Ticket, incident, problem and change transitions.
Security	Finding, privileged action, break-glass and review.
Administration	Role, configuration and retention change.

ID	Area	Requirement	Priority	Acceptance signal
IT-AUD-001	Actor	Audit shall distinguish actual and effective actor.	Must	Impersonation/support context is visible.
IT-AUD-002	Secrets	Audit shall never contain raw secret material.	Must	Automated scan passes.
IT-AUD-003	Retention	IT records shall follow configurable retention and legal hold.	Must	Held record cannot be purged.
IT-AUD-004	Integrity	IT users cannot alter or permanently delete audit.	Must	Database/API controls enforce append-only policy.

52. Search, Reporting and Analytics

52.1 Search domains

- Applications and integrations
- Provider connections
- Users and account mappings
- Devices

- Software and licences
- Tickets and incidents
- Problems and changes
- Configuration items
- Knowledge articles
- Security findings and access reviews

52.2 Metrics

Category	Examples
Provisioning	Success, failure, duration and backlog.
Integration	Availability, failure rate, sync delay and conflicts.
Licences	Utilization, unused, overallocated and renewal cost.
Devices	Compliance, age, lost and unsupported.
Service Desk	Volume, response, resolution, first-contact resolution and SLA.
Incident	Major incidents, restoration time and recurrence.
Change	Success, failure, rollback and emergency percentage.
Security	MFA, findings, privileged accounts and review completion.
Financial	Permission-safe IT spend and renewal forecast.

ID	Area	Requirement	Priority	Acceptance signal
IT-REP-001	Permission	Search and analytics shall apply row and field authorization before aggregation.	Must	No count or field leakage.
IT-REP-002	Freshness	Derived metrics shall show data time and source.	Must	Stale metric is labelled.
IT-REP-003	Privacy	Reports shall not include email content, private event detail or secrets.	Must	Export tests pass.
IT-REP-004	Trace	Metric shall drill to authorized source records.	Should	No editable analytical master exists.

53. Observability, Failure Handling and Operations

Signal	Purpose
Connection health	Provider, credential, consent and capability state.
Token expiry	Prevent authentication outage.
Webhook failure	Detect missed near-real-time changes.
Polling lag	Detect delayed provider state.
Queue backlog	Detect blocked synchronization.
Rate-limit state	Control backoff and operator expectations.
Mapping conflict	Surface unsafe identity/resource ambiguity.
Dead letters	Require technical or business review.

Signal	Purpose
Provisioning failure	Prevent incomplete joiner/offboarding.
Device revocation delay	Protect lost/terminated-user devices.
Ticket SLA	Operational service performance.
Service account use	Detect dormant or anomalous automation.

Failure class	Retry behavior
Network/provider temporary	Exponential backoff with jitter.
Rate limited	Respect provider retry window.
Credential expired	Attempt allowed refresh; otherwise suspend.
Consent revoked	Do not retry; require reauthorization.
Scope insufficient	Do not retry; review required.
Mapping ambiguous	Do not retry; human resolution.
Version conflict	Create conflict or refresh source.
Security violation	Quarantine/suspend and alert.
Validation permanent	Return deterministic failure.

ID	Area	Requirement	Priority	Acceptance signal
IT-OPS-001	Runbooks	Every critical adapter and queue shall have owner and runbook.	Must	Operational registry contains current runbook.
IT-OPS-002	Tracing	Cross-service calls shall preserve trace and correlation IDs.	Must	One trace shows command, provider and consumer hops.
IT-OPS-003	Dead letter	Replay shall recheck current scope, state and permissions.	Must	Obsolete effect is not recreated.
IT-OPS-004	Circuit breaker	Provider failure shall be isolated from core platform.	Must	Other modules remain operational.

54. Non-Functional Requirements

Area	Requirement
Availability	99.9% target for core IT administration; provider dependencies reported separately.
Performance	Common IT lists p95 <= 2 s; dashboard p95 <= 2.5 s under target load.
Authorization	Cached common decision p95 <= 250 ms; sensitive actions may use live evaluation.
Scalability	Thousands of companies, multiple providers and millions of mappings.
Reliability	Durable queues, idempotency, retry, dead letter and reconciliation.
Security	MFA for privileged roles, encryption, secrets manager and penetration testing.
Accessibility	WCAG 2.1 AA target for core workflows.
Localization	English and Albanian resources with locale-safe dates/time.

Area	Requirement
Audit	Immutable privileged-action evidence.
Recovery	Documented RPO/RTO and tested restoration for IT records.
Compatibility	Provider adapter versions and deprecation registry.
Maintainability	No tenant-specific code forks; configuration and feature flags.

55. Testing Strategy and Critical Scenarios

55.1 Test layers

- Unit tests for state machines, mapping, scopes, SLA and policy
- Contract tests for every provider adapter
- Database tests for tenant/company constraints and idempotency
- Integration tests for secrets, queues, webhooks, retry and reconciliation
- Security tests for scopes, token handling, SSRF, replay and cross-company isolation
- End-to-end tests for JML, calendar, email, devices, licences and Service Desk
- Resilience tests for provider outage, rate limit and credential expiry
- Accessibility and mobile responsive tests
- Migration and rollback tests

55.2 Critical scenarios

29. IT Admin connects Microsoft without required approval.
30. IT Admin attempts to read employee mailbox content.
31. Outlook sends the same event twice.
32. External event conflicts with issued ConstructionOS meeting.
33. Provider consent is revoked externally.
34. User removed from provider but remains an active employee.
35. HR terminates employee while device is offline.
36. Account mapping finds two possible users.
37. Provider scope expands after re-consent.
38. Shared mailbox mapped to wrong company.
39. Helpdesk Agent attempts to grant Company Admin.
40. Lost device contains offline workspace.
41. Remote wipe requested without approval.
42. Service account expires during synchronization.
43. Provider rate limit is reached.
44. Webhook signature is invalid or replayed.
45. Licence allocation exceeds entitlement.
46. Intercompany transfer recalculates provider and access state.
47. Platform Admin attempts tenant access without Support Session.
48. Integration revoked with pending jobs.
49. Search exposes provider-secret metadata.
50. Audit receives token value.
51. SSO provider fails during mandatory mode.
52. Company deletion standby blocks IT mutations.
53. Reconciliation detects orphan external mapping.
54. Business-domain permission is absent despite IT administrator role.
55. Emergency change completes without post-review.
56. Security incident visibility leaks to general Helpdesk.

56. Acceptance Criteria

- Every company receives a protected IT Department and baseline technical teams.
- Company IT and Platform Admin authority are technically separated.

- IT membership does not grant unrestricted IT or business capabilities.
- IT cannot read mailbox content, private calendar detail, Payroll, Legal or Finance by default.
- Microsoft 365 and Google Workspace use provider-neutral contracts.
- ConstructionOS core functions remain available during provider outage.
- SSO, domains, account mappings and provisioning are versioned and auditable.
- HR triggers JML but Authorization remains authoritative for business permission.
- Offboarding revokes access while preserving historical records.
- Devices reference Assets and do not duplicate the physical master.
- Licences link to Procurement, Contracts and Finance without duplicate commercial truth.
- Operational modules route email delivery through Notifications.
- Calendar synchronization preserves source ownership and conflict rules.
- Connected Applications require named technical, business and data owners.
- Scope expansion requires review and cannot activate silently.
- Secrets are stored only through secret references.
- Synchronization is idempotent, version-aware and reconcilable.
- Service Desk supports requests, incidents, problems, changes, SLA and knowledge.
- Security findings and privileged access reviews are independently governed.
- Permission revocation propagates to sessions, providers, devices and mobile workspaces.
- Every privileged IT action is immutable-audited without secret values.
- All critical tenant/company isolation and provider-security tests pass.

57. Delivery Phases and Rollout

Phase	Scope	Exit criteria
1 - IT Foundation	Protected department, roles, dashboard, Service Desk, device/licence registry and audit.	Company IT can operate core internal workflows.
2 - Microsoft 365	Domains, tenant connection, user mapping, Outlook, rooms, Teams and email delivery.	Pilot company completes security and sync certification.
3 - Google & Adapter Framework	Google organization, Calendar, Gmail, Meet and generic provider runtime.	Two providers use shared contracts.
4 - Enterprise Identity	SSO enforcement, directory sync, automated provisioning, service accounts and access reviews.	JML and privileged governance certified.
5 - ITSM & Security Maturity	Problems, changes, CMDB, major incidents and security operations.	Operational metrics and runbooks approved.
6 - Advanced Integrations	Endpoint, storage, backup, network/site and security-provider adapters.	Each adapter independently certified.

58. Migration and Adoption Strategy

57. Inventory existing administrator accounts, providers, domains, devices, licences and integrations.
58. Create Connected Application records and assign owners.
59. Map existing users/provider accounts through staged duplicate review.
60. Import licences and device technical profiles with Assets links.
61. Configure email/calendar providers in test mode.
62. Pilot one company and one department before mandatory SSO or automated provisioning.
63. Run parallel reconciliation before switching synchronization authority.
64. Train Company Owner, IT, Helpdesk, HR and managers on boundaries.
65. Retire spreadsheets and undocumented shared credentials only after validation.
66. Complete access review and remove legacy excessive permissions.

ID	Area	Requirement	Priority	Acceptance signal
IT-MIG-001	Staging	Imports shall use validation, preview and controlled commit.	Must	No direct uncontrolled table write.

ID	Area	Requirement	Priority	Acceptance signal
IT-MIG-002	Parallel run	Critical provider sync shall reconcile before cutover.	Must	Mismatch threshold meets sign-off.
IT-MIG-003	Rollback	SSO/provider cutover shall include tested rollback/fallback.	Must	Pilot recovery test passes.
IT-MIG-004	Evidence	Migration decisions, mappings and exceptions shall be audited.	Must	Post-migration report is complete.

59. Risks and Open Decisions

Topic	Decision / risk requiring resolution
Microsoft permissions	Exact initial scopes and application/delegated split.
Google permissions	Exact initial scopes and domain-wide delegation policy.
SSO protocols	Initial SAML/OIDC provider support and emergency fallback.
Provisioning protocol	Whether SCIM or provider-native provisioning is first.
Email content	Whether any future mailbox-content use case is allowed and by whom.
File connectors	Timing and scope for SharePoint, OneDrive and Google Drive.
Endpoint management	Native capability vs external MDM adapters.
Remote wipe	Legal/ownership policy for personal or mixed-use devices.
CMDB depth	Initial manual/imported inventory vs discovery integrations.
Project/site network	Initial functional depth and supported devices.
Data residency	Provider-region and derived-data constraints.
Retention	Ticket, integration log and security evidence periods.
Licensing	Which external portal/guest identities consume IT-managed licences.
Group IT	Default parent-company authority and child-company restrictions.
Support model	Internal IT only vs approved managed-service provider portal.

60. Definition of Done

- All Must requirements and acceptance criteria are implemented and tested.
- Domain ownership is enforced in database, API, events, projections and UI.
- Company IT cannot obtain business-data access from technical role alone.
- Protected IT Department provisioning is migration-safe and repeatable.
- Microsoft and Google adapters pass contract, security, idempotency and resilience tests.
- Secrets manager, rotation, redaction and access controls are production-ready.
- JML flows coordinate HR, Identity, Authorization, Assets, Mobile and providers.
- Service Desk, incident, problem and change lifecycles are auditable.
- Devices and licences reconcile with Assets and commercial source records.
- Observability dashboards, alerts, queues, dead letters and runbooks are approved.
- Backup, recovery, migration and provider-revocation tests pass.
- No unresolved critical security, privacy, isolation or data-integrity defect remains.

Final product position

IT Administration, Integrations & Service Management is the controlled technical operating layer for each ConstructionOS company. It enables systems and access without becoming a competing source of business truth or a universal business-data administrator.

61. Page and Flow Registry

Code	Page / flow	Primary purpose
IT-001	IT Dashboard	Company technical health and action overview.
IT-002	People & Access	User technical profile, mappings and JML.
IT-003	Identity Providers	SSO, domains, provisioning and MFA.
IT-004	Microsoft 365	Tenant, Outlook, Teams, rooms and mailboxes.
IT-005	Google Workspace	Directory, Calendar, Gmail and Meet.
IT-006	Connected Applications	Application registry and governance.
IT-007	Integration Detail	Scopes, consent, mappings, sync, failures and audit.
IT-008	Integration Request Wizard	Business, data, security and technical review.
IT-009	Integration Monitoring	Jobs, rate limits, failures, dead letters and reconciliation.
IT-010	Devices	Technical profile, compliance and actions.
IT-011	Software & Licences	Pools, allocation, usage and renewals.
IT-012	Service Catalogue	Employee-requestable IT services.
IT-013	My IT Requests	Employee request tracking.
IT-014	IT Ticket Workspace	Triage, assignment, SLA and resolution.
IT-015	Major Incident Workspace	Command, timeline, communication and restoration.
IT-016	Problem Workspace	Root cause, known error and corrective change.
IT-017	Change Workspace	Assessment, approval, implementation and rollback.
IT-018	CMDB	Configuration items and dependencies.
IT-019	Security Operations	Findings, incidents and privileged activity.
IT-020	Access Reviews	Campaigns, decisions and certification.
IT-021	Service Accounts	Automated identities and credential lifecycle.
IT-022	Knowledge Base	Approved technical guidance.
IT-023	IT Audit	Privileged technical activity and evidence.
IT-024	IT Settings	Company-level technical configuration.

Appendix A - Core Capability Catalogue

Capability	Purpose
it.department.view	View IT workspace and permitted summaries.
it.department.manage	Manage IT department configuration.
it.users.view	View technical identity profile.

Capability	Purpose
it.users.provision	Create/link provider account through approved process.
it.users.suspend	Suspend technical account.
it.users.revoke_sessions	Revoke active sessions.
it.users.map_external_account	Manage provider account mapping.
it.identity_providers.view	View provider configuration/health.
it.identity_providers.configure	Create/edit provider draft.
it.identity_providers.test	Execute safe connection test.
it.identity_providers.activate	Activate approved provider.
it.identity_providers.revoke	Revoke provider connection.
it.integrations.view	View connected applications and connections.
it.integrations.request	Request new integration.
it.integrations.configure	Configure approved connection.
it.integrations.test	Run connection/capability test.
it.integrations.activate	Activate after approval.
it.integrations.suspend	Suspend new operations.
it.integrations.revoke	Permanently revoke connection.
it.integrations.reconcile	Run source-target reconciliation.
it.integration_scopes.review	Review requested scopes.
it.integration_scopes.approve	Approve provider scopes.
it.devices.view	View technical device profiles.
it.devices.manage	Manage device technical configuration.
it.devices.revoke	Revoke device access.
it.devices.remote_wipe_request	Request remote wipe.
it.devices.remote_wipe_approve	Approve remote wipe.
it.licences.view	View licence pools and assignments.
it.licences.manage	Manage products/pools/renewals.
it.licences.assign	Assign entitlement.
it.licences.recover	Recover entitlement.
it.service_desk.create	Create service record.
it.service_desk.assign	Assign IT ticket.
it.service_desk.resolve	Resolve ticket.
it.incidents.manage	Manage incidents/major incidents.
it.problems.manage	Manage problem/known error.
it.changes.manage	Prepare and execute changes.
it.changes.approve	Approve eligible changes.
it.security.view	View permitted security operations.

Capability	Purpose
it.security.manage	Manage findings/incidents.
it.break_glass.activate	Activate emergency access.
it.break_glass.review	Perform mandatory post-review.
it.access_reviews.manage	Manage access-review campaign.
it.audit.view	View IT audit evidence.
it.configure	Manage company IT settings.

Appendix B - Standard Role Catalogue

Role template	Default scope and intent
IT Director	Company/group strategic IT governance.
IT Manager	Company IT operations and team oversight.
System Administrator	Company technical identity and system settings.
Identity & Access Administrator	Company identity, mappings, provisioning and access reviews.
Integration Administrator	Approved provider connections and synchronization.
Security Administrator	Security findings, privileged controls and incidents.
Device Administrator	Endpoint technical profiles and revocation.
Software & Licence Manager	Company software entitlements and renewals.
Helpdesk Agent	Assigned Service Desk records and ordinary recovery.
IT Change Manager	Change assessment, scheduling and governance.
Major Incident Manager	Major incident command and review.
IT Auditor	Read-only company IT evidence.
Project/Site IT Administrator	Explicit assigned project/location.
Employee IT Self-Service	Own requests and published knowledge.

Appendix C - Reference Integration Profiles

Profile	Default capabilities	Key restrictions
Microsoft Calendar - Free/Busy	Account mapping, availability and selected resources.	No private title, description or attendee detail.
Microsoft Calendar - Managed	Selected event read/write under explicit authority.	Conflict review and no issued-record overwrite.
Microsoft Notification Email	Send approved notifications through company identity.	No mailbox reading.
Google Calendar - Free/Busy	Availability and resource mapping.	No private event detail.
Google Calendar - Managed	Selected calendar synchronization.	Version/authority mapping required.
Generic SMTP	Notification sending only.	Credentials isolated; no inbox access.
Directory Read	Selected user metadata and status.	No automatic business-role grant.
Directory Provisioning	Create/disable provider accounts.	Requires workflow, reconciliation and protected scope.

Profile	Default capabilities	Key restrictions
File Connector - Future	Selected library/folder metadata and file operations.	Documents remains authoritative; restricted classes excluded.
Endpoint Adapter - Future	Compliance, app version, revoke/wipe command.	Assets remains physical source; approval for destructive action.